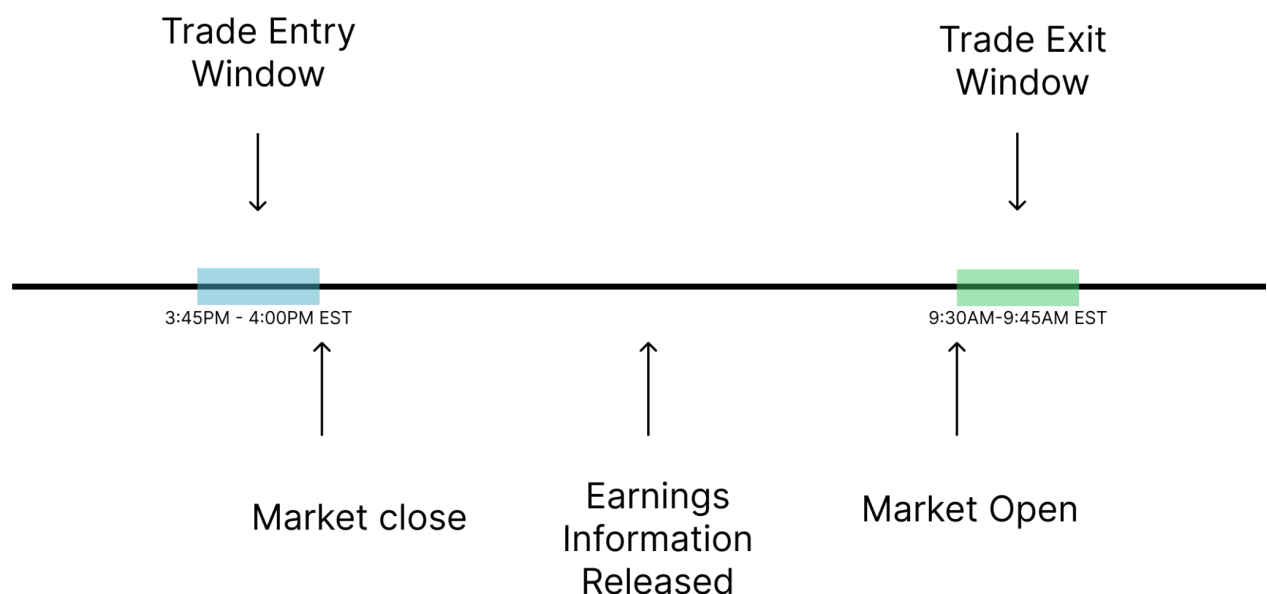


How to structure a trade to capture risk premiums around earnings events

The quick answer:

- Sell an at-the-money straddle on the closest expiration date.
- Open your trade right before market close (before the event).
- Close your trade 5-10 min after the market open (after the event).



The long answer:

The most important thing to remember is that the trade structure we pick is literally just the tool we use to express our view on a ticker.

Need to tighten a screw? Use a screwdriver. Need to cut a piece of wood? Use a saw.

Picking a trade structure is no different.

So what job are we trying to complete with the Earnings Premium strategy? *What view are we trying to express?* Let's go through a couple of questions to figure it out.

Do we have a view on...

1. If the stock goes up or down? **No, we don't know/care which direction it goes.**
2. If the stock moves more or less than implied? **Yes, we think it will move less.**
3. If implied volatility goes up or down? **Yes we think it will move down.**
4. What is our time horizon for the trade? **Less than 1 day.**

Based on these answers, we know that we want a structure that has the following characteristics

- **Delta neutral** (no PnL impact from direction)
- **Long theta** (make money as time passes)
- **Short gamma** (lose money if big moves)
- **Short vega** (make money if IV crushes)
- **Maximum exposure to tomorrow's outcome**

The structure that is the best fit for the job we are looking to accomplish is a **short straddle with a close expiration date**.

Let's talk about the decisions we've made when choosing this structure.

Why we chose straddles over strangles

Both straddles and strangles fit the "structure profile" we are looking for with this strategy. When we talked about the ETF VRP strategy, I shared with you my reasoning for why I think strangles are the optimal structure *especially for retail traders*.

But for earnings, I will actually be sharing why straddles are the superior structure.

The logic is simple.

Since we don't have a need to delta hedge because of our short time horizon, there is less added benefit to trading strangles.

Trading the straddle allows us to receive better feedback on if our view that implied volatility should be higher than realized volatility is making us money. This works because the straddle price is roughly equal to the implied move, with no added buffer from the wider strikes.

So even though the strangle gives us wider break-evens, it doesn't actually impact our long term EV in any way and it actually makes it harder for us to get feedback when running the strategy about if what we are seeing is in line with what we should expect and if we are doing a good job with our trade execution (mistakes are more obvious).

Why do we want to trade a close expiration?

We want to trade the closest to 1 DTE expiration available on the option chain. The logic behind this is the same as the reason why we open before the market closes the day before the event and close out in the morning after the event.

We only have a view on the earnings event and want to isolate our exposure to it.

By trading a shorter DTE, the option price is less influenced by the “non earnings event” days.

Think about it. If earnings are tomorrow and the option has 1 day to expiration, it expires very shortly after the event, meaning that the event plays a big role in the value of the option.

But if the option we trade has *1 year* to expiration, all of a sudden there are hundreds of “regular days” that dilute the impact of the earnings event on this option. There is just not enough exposure to the difference between the implied and realized move for us to see a meaningful PnL.

So that's why the shorter the DTE, the better for this strategy.

Why don't we buy wings or “protection”? Are there better ways to hedge our risk?

At Predicting Alpha we are **not** a fan of buying wings. Our core reason is quite simple:

We get paid for holding the risk of a big move happening. If we hedge away the risk of a big move happening for ourselves, why are we getting paid?

Remember: We get paid for providing a service to the market. *If we no longer offer the service, we no longer get paid.*

“But Sean, what about if the stock moves a crazy amount?”

Well, there are a couple things I have to say about that.

First of all, you will take a big hit. That is the nature of being short volatility and we need to embrace the fact that we will inevitably experience situations like that sometimes. Once again, *it's the reason we are getting paid.*

Secondly, the way that you should be hedging your risk is actually through diversification. Remember, the beauty with earnings trading is that you are able to trade *literally hundreds of events each season*, which means:

- a) We can spread our risk across a lot of bets
- b) We are not counting on one trade to make or break our PnL.

If one big loser takes you out of the game, you either sized up way too much or you are trading a ticker that is way too big for your account size.

If you absolutely insist on buying wings..

You should look at buying them extremely far out of the money. To the point where they only cost a couple pennies. Because when you are this far out, even though you are paying a lot in implied volatility terms, you are only paying a few pennies.

My recommendation is to place them *at least* 2x the straddle price away.

For example, If the stock price is \$100 and the straddle is going for \$5, the closest you should be buying wings is at \$90 and \$110.

Stop thinking about risk/reward ratios

These wings are not in place to “give you a good risk reward ratio”. They are there to hedge away the absolute worst case scenario and nothing else.

You don't even close these out in the morning, you just let them expire worthless (hopefully every single time!).

In my opinion this is the only acceptable way to buy wings. Everything else is risk aversion to the point where it's not even worth trading volatility, so focus on risk management through diversification and if you absolutely need wings, buy them very far out of the money.

Let's look at a real example of how wings impact your trade

This is what your payoff graph looks like when you sell a naked straddle.



Since we are trading a very short DTE (in this case it's 2 DTE), the price of the straddle is mostly made up with the implied move from the earnings event.

For this reason, the breakevens we see cover a similar size move up or down as is the implied move for the event.

This trade is purely the implied VS realized move. Even though the risk of extremely large moves can be scary, this is the most direct way to trade the implied VS realized move.

The picture below is what our payoff graph looks like if we decide to purchase wings. In this example we are buying the wings 2x the straddle price away from at the money (as we suggested above).



As you can see, when we purchase the wings our max profit decreased from \$218.50 to \$203.50. This is about a 7% decrease in the option premium we collect.

Now at first you might think that this is not a big deal.

We can get protection and still collect 93% of the premium! Isn't this a no brainer?

Slow down for a second. Let's think this through.

Buying wings is a no brainer *if the entire straddle is supposed to be worthless.*

But it isn't. The VRP embedded isn't the entire option price. It's the *excess price above fair value.*

A reasonable assumption here would be to say that the excess is maybe on average 20%.

Which means that our average return just went down from 20% to 13% because we bought the wings. **It's a 35% reduction in our expected value!**

These wings are *far* from cheap. Just like if you choose to purchase the long strangle hedge in the ETF Premium strategy, you should expect a meaningful decrease in your absolute returns. After all, there's no free money in the market, and if someone is going to hold your "worst case" risk for you, you should expect to pay.

You could buy wings, but really just trade the straddle and size down

If you feel uncomfortable with the exposure of the straddle, just trade smaller and trade more tickers. You don't need to be leveraged to the maximum from day 1, you just need to be making smart decisions consistently.

By trading smaller sizes on each individual trade, it also frees up capital for you to put on even more trades (which is better for the expectancy of the strategy!).

It's the home stretch. The final chapter for the Earnings Premium Strategy. Let's talk about how to manage these trades.